

CIS — Nova 1 Chain 22016 v1.0

Component Integration Specification

Field	Value
Document ID	CIS-NOVA-1-CHAIN-22016-v1
Version	1.0
Status	Draft
Chain string ID	nova-1
Network ID (decimal)	22016
Network ID (hex)	0x5600
Chain type	evm (EVM-compatible)
Native symbol	NOVA

1. Purpose and scope

This CIS defines how **Nova 1 Chain** (network ID **22016**) integrates with the OneX bridge, token platform, ledger settlement middleware, and Nova Bank Online.

In scope

- Chain registry entry and MetaMask network parameters
- JSON-RPC connectivity and explorer
- Ledger settlement: fiat → Nova 1 on-chain payout
- Token platform deploy and cross-chain wrap
- HYBX exchange route `hybx-nova-1`
- Environment and deployment contract

Out of scope

- Nova 1 consensus node operation (assumes external RPC operator)
- Smart contract audit
- Nova Bank Online account management (see `CIS-Nova-Bank-Online-v1.md`)

2. Network parameters

2.1 Chain registry (`configs/chains.json`)

```
{
  "id": "nova-1",
  "name": "Nova 1 Chain",
  "networkId": 22016,
  "symbol": "NOVA",
  "native": false,
  "rpc": "https://rpc.nova1.chain",
  "explorer": "https://explorer.nova1.chain",
  "color": "#6b4ce6",
  "type": "evm"
}
```

2.2 MetaMask / wallet network table

Field	Value
Network name	Nova 1 Chain
Chain ID	22016
Chain ID (hex)	0x5600
Currency symbol	NOVA
RPC URL	https://rpc.nova1.chain (replace with live endpoint)
Block explorer	https://explorer.nova1.chain
OneX string ID	nova-1

2.3 Chain aliases (OneX bridge)

Alias	Resolves to
nova	nova-1
nova-1	nova-1
nova1	nova-1
22016	nova-1
0x5600	nova-1

External destination format: nova-1:0xRecipientAddress

3. Architecture

[Architecture diagram — see markdown source]

```

flowchart LR
    subgraph nova [Nova 1 Chain 22016]
        RPC[JSON-RPC]
        EXP[Block explorer]
        NOVA[NOVA native gas]
    end

    subgraph onex [OneX Bridge]
        CH[Chain adapter evm]
        TP[Token platform]
        SET[Ledger settlement]
        HY[HYBX middleware]
    end

    subgraph bank [Nova Bank Online]
        OB[Online bank / NSB]
        LG[Real ledger M0/M1/NSB]
    end

    LG -->|POST /bridge/ledger/settle| SET
    SET -->|ONEX_EVM_SENDER_KEY| CH
    CH --> RPC
    HY -->|hybx-nova-1| CH
    TP -->|deploy/wrap| CH
    OB --> LG

```

4. Environment matrix

Variable	Purpose	Example
ONEX_DEFAULT_BRIDGE_CHAIN	Default ledger bridge target	nova-1
NOVA1_RPC_URL	Nova 1 JSON-RPC endpoint	https://rpc.nova1.chain
NOVA1_EXPLORER	Block explorer base URL	https://explorer.nova1.chain
NOVA1_CHAIN_ID	Decimal chain ID	22016
ONEX_EVM HOLDER	Read-balance treasury address	0x...
ONEX_EVM_SENDER_KEY	Settlement signer (64-hex key)	<private-key>
ONEX_LEDGER_MODE	Ledger mode	production
ONEX_API_KEY	Bridge API authentication	<secret>

Template: `deploy/env.nova-1-22016.example`

```
ONEX_LEDGER_MODE=production
ONEX_DEFAULT_BRIDGE_CHAIN=nova-1
NOVA1_RPC_URL=https://rpc.nova1.chain
NOVA1_EXPLORER=https://explorer.nova1.chain
NOVA1_CHAIN_ID=22016
ONEX_EVM HOLDER=0xYourNova1TreasuryAddress
# ONEX_EVM_SENDER_KEY=64hexPrivateKeyWithNOVAForGas
```

5. API contract

5.1 Ledger settlement (fiat → Nova 1)

Endpoint: `POST /bridge/ledger/settle`

```
{
  "fromAccount": "nova-usd-checking",
  "amount": "500",
  "payoutAsset": "NOVA",
  "kind": "real_crypto",
  "externalTo": "nova-1:0xRecipient0nNova1Chain"
}
```

Preview (no on-chain tx):

```
{
  "fromAccount": "m1-usd-checking",
  "amount": "100",
  "payoutAsset": "NOVA",
  "kind": "real_crypto",
  "externalTo": "nova-1:0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb0",
  "preview": true
}
```

5.2 Ledger status

```
curl -s https://HOST/bridge/ledger/status | jq '.defaultBridgeChain'
# Expected: "nova-1"
```

5.3 Token platform

Method	Path	Nova 1 usage
GET	/bridge/platform/status	Platform health
POST	/bridge/platform/deploy	Deploy ERC-20 on Nova 1
POST	/bridge/platform/wrap	Wrap to/from Nova 1
GET	/bridge/platform/tokens	List tokens including Nova 1

Deploy example:

```
POST /bridge/platform/deploy
{
  "name": "Nova Stable",
  "symbol": "NUSD",
  "supply": "1000000",
  "chain": "nova-1",
  "decimals": 18
}
```

5.4 HYBX exchange route

Add route `hybx-nova-1` to HYBX middleware config (`configs/hybx-middleware.example.json`):

```
{
  "routes": [
    "nsb-hybx",
    "hybx-nova-1",
    "nova-1-hybx"
  ]
}
```

Method	Endpoint	Purpose
GET	/bridge/bank/hybx/exchange/routes	List routes including Nova 1
POST	/bridge/bank/hybx/exchange	Exchange Nova Bank ↔ Nova 1

5.5 Production status

```
curl -s https://HOST/bridge/production/status | jq '.ledger.defaultBridgeChain'
```

6. JSON-RPC

Nova 1 Chain 22016 exposes standard EVM JSON-RPC. OneX uses the `evm` chain adapter.

Method	Returns
<code>eth_chainId</code>	<code>0x5600</code>
<code>eth_blockNumber</code>	Latest block
<code>eth_getBalance</code>	NOVA balance
<code>eth_sendRawTransaction</code>	Settlement tx hash

RPC resolution order:

1. `NOVA1_RPC_URL` environment variable
2. `configs/chains.json` → `nova-1.rpc` fallback

7. Security

Control	Requirement
Sender key	<code>ONEX_EVM_SENDER_KEY</code> — fund with NOVA for gas; never commit
Treasury holder	<code>ONEX_EVM_HOLDER</code> — read-only balance queries
RPC TLS	HTTPS required in production
Chain ID validation	Adapter must confirm <code>22016</code> before signing

8. Deployment

Deploy guide: `deploy/DEPLOY-nova-1-22016.md`

8.1 Quick deploy

```
ssh ubuntu@YOUR_NOVA1_BRIDGE_SERVER
git clone https://github.com/zaragoza444/onex.git ~/onex
cd ~/onex
cp deploy/env.nova-1-22016.example /etc/onex/onex.env
# Edit ONEX_API_KEY, ONEX_EVM_SENDER_KEY, NOVA1_RPC_URL
sudo systemctl restart onex-bridge
```

8.2 Post-deploy verification

```
curl -s http://127.0.0.1:9338/bridge/ledger/status | jq '.defaultBridgeChain, .dbis138Configured'
curl -s http://127.0.0.1:9338/bridge/platform/status | jq .
```

Wallet: `http://HOST:9338/wallet/#ledger` → default bridge tab: **Nova 1 Chain**

8.3 Acceptance criteria

- `defaultBridgeChain` = `"nova-1"`
- `nova-1` appears in `configs/chains.json` and supported chains API
- Settlement preview succeeds for `externalTo: "nova-1:0x..."`
- Live settlement submits tx when sender key funded with NOVA
- Token deploy on `nova-1` returns contract address

9. Supported external destination examples

Format	Example
Short	<code>nova-1:0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb0</code>
Alias	<code>nova:0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb0</code>
Chain prefix	<code>chain:nova-1:0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb0</code>

10. Related documents

Document	Path
Nova Bank Online CIS	docs/cis/CIS-Nova-Bank-Online-v1.md
Integration matrix	docs/cis/CIS-Nova-Integration-Matrix-v1.md
DBIS-138 pattern (reference)	deploy/DEPLOY-dbis-138.md
Token platform API	docs/TOKEN-PLATFORM.md
Chain registry	configs/chains.json